1 Metadata Input (Table Schema)

Column	Description
Order ID	Unique identifier
Product ID	Item identifier
Unit price	Price per item
Quantity	Number of units
Net amount	Total value
Supplier number	Supplier code
…	…

2 LLM-based Schema Reasoning

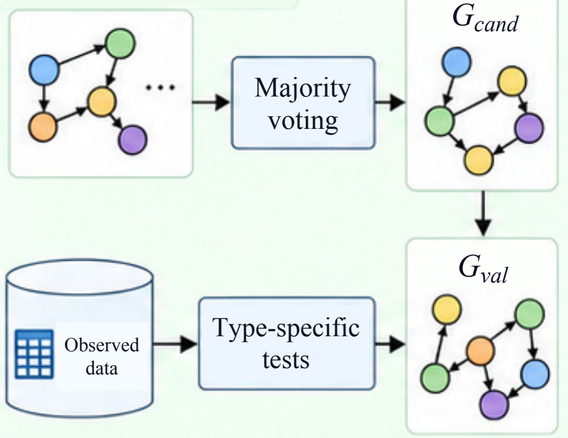
LLM A
Net amount = Unit price × Quantity
Supplier number → Supplier name
…

LLM B
Order ID → Order date
Category → Price range
…

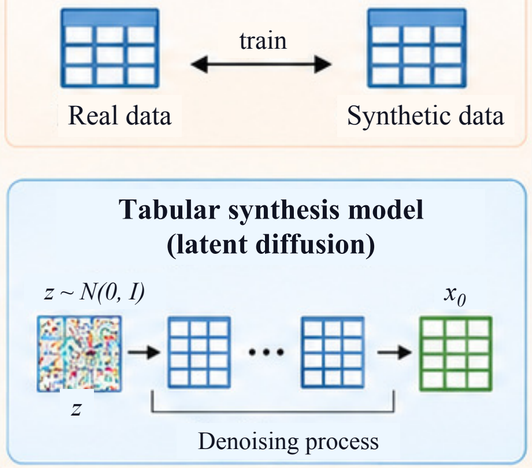
LLM C
Quantity → Stock level
Product category → Category code
…

3 CR-KG Construction and Validation

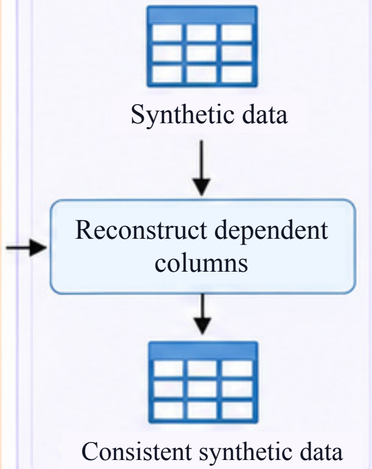
Candidate relationships



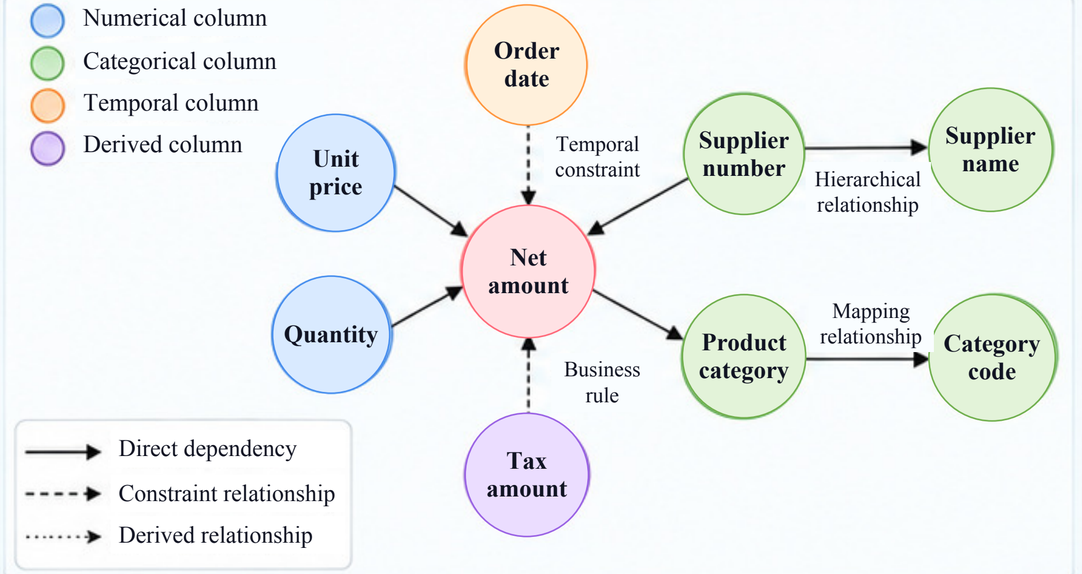
4 Train Tabular Synthesis Model



5 Generate Consistent Synthetic Data



Example Validated Column Relationship Graph



Learned Relationship Examples

Mathematical relationship (joint parent set)
Net amount = Unit price × Quantity
Parents: {Unit price, Quantity}
Type: mathematical

Hierarchical relationship
Supplier number → Supplier name
Parent: {Supplier number}
Type: hierarchical

Mapping relationship
Product category → Category code
Parent: {Product category}
Type: mapping

Observed Data and Type-Specific Tests

Order ID	Unit price	Quantity	Net amount
A001	10.0	3	30.0
A002	25.5	2	51.0
A003	8.0	5	40.0
…	…	…	…

Type-specific tests

- Numerical consistency (e.g., equation test)
- Categorical validity (e.g., value matching)
- Temporal consistency (e.g., date ordering)
- Distributional plausibility (e.g., range check)